



# Project Report

## Heart Disease Prediction Using K-Nearest Neighbours (KNN) Classification

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## 1. Introduction

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Cardiovascular disease remains one of the leading causes of mortality worldwide, responsible for an estimated 17.9 million deaths each year according to the World Health Organisation. Heart disease, a subset of cardiovascular conditions, is particularly prevalent and costly, both in terms of human lives and healthcare expenditure. Early and accurate prediction of heart disease risk can enable timely medical intervention and significantly improve patient outcomes.

Machine learning offers a compelling approach to this prediction problem by learning decision boundaries from historical patient data without explicitly programming disease-specific rules. Among classical machine learning algorithms, the K-Nearest Neighbours (KNN) classifier is notable for its simplicity, interpretability, and instance-based learning paradigm—making it a useful baseline model for binary medical classification tasks.

This project applies a KNN classifier to a publicly available heart disease dataset containing records for 10,000 patients across 21 clinical and lifestyle features. The work encompasses the complete machine learning pipeline: data exploration, missing-value imputation, categorical encoding, feature scaling, model training, and evaluation. The primary goal is to build a working predictive system and analyse its behaviour, with a critical discussion of the model's limitations—particularly the impact of class imbalance on minority-class recall.

The report is organised as follows. Section 2 describes the dataset and its features. Section 3 covers exploratory data analysis. Section 4 details the preprocessing pipeline. Section 5 introduces the KNN algorithm. Section 6 reports evaluation results. Section 7 discusses findings and limitations. Section 8 concludes the report.

## 2. Dataset Description

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### 2.1 Overview

The dataset used in this project is the *Heart Disease* dataset sourced from Kaggle (user: oktayrdeki). It comprises **10,000 patient records** and **21 attributes**, including both clinical measurements and lifestyle indicators. The target variable is **Heart Disease Status** (binary: *Yes / No*), making this a supervised binary classification problem.

The dataset exhibits a class imbalance: approximately 80% of records correspond to

patients without heart disease and 20% to patients with heart disease.

2.2 Feature Description

The 21 features are divided into two groups based on their data type:

2.2.1 Numerical Features

Table 1: Numerical Features in the Heart Disease Dataset

| Feature             |  | Description                                      |
|---------------------|--|--------------------------------------------------|
| Age                 |  | Patient age in years (range: 18–80+)             |
| Blood Pressure      |  | Systolic blood pressure in mmHg (range: 120–180) |
| Cholesterol Level   |  | Total cholesterol in mg/dL (range: 150–300)      |
| BMI                 |  | Body Mass Index (range: 18–50)                   |
| Sleep Hours         |  | Average daily sleep duration (hours)             |
| Triglyceride Level  |  | Triglyceride concentration in mg/dL              |
| Fasting Blood Sugar |  | Fasting glucose level in mg/dL                   |
| CRP Level           |  | C-Reactive Protein level (inflammation marker)   |
| Homocysteine Level  |  | Plasma homocysteine in $\mu\text{mol/L}$         |

2.2.2 Categorical Features

Table 2: Categorical Features in the Heart Disease Dataset

| Feature              | Description / Categories  |
|----------------------|---------------------------|
| Gender               | Male / Female             |
| Exercise Habits      | Low / Medium / High       |
| Smoking              | Yes / No                  |
| Family Heart Disease | Yes / No (family history) |
| Diabetes             | Yes / No                  |
| High Blood Pressure  | Yes / No                  |
| Low HDL Cholesterol  | Yes / No                  |
| High LDL Cholesterol | Yes / No                  |
| Alcohol Consumption  | Low / Medium / High       |
| Stress Level         | Low / Medium / High       |
| Sugar Consumption    | Low / Medium / High       |
| Heart Disease Status | Target: Yes / No          |

3. Exploratory Data Analysis

3.1 Initial Data Inspection

The first five rows of the dataset were inspected to confirm correct loading and understand the general structure. The dataset contains 10,000 rows and 21 columns. Detailed type information confirms that numerical columns are stored as `float64` and categorical columns as `object`. No duplicate rows were found in the dataset.

3.2 Missing Value Analysis

An initial null-value audit revealed missing entries in every column. Table 3 summarises the counts.

**Table 3:** Missing Value Counts Before Imputation

| Feature                    | Missing Count | Missing %     |
|----------------------------|---------------|---------------|
| Age                        | 29            | 0.29%         |
| Gender                     | 19            | 0.19%         |
| Blood Pressure             | 19            | 0.19%         |
| Cholesterol Level          | 30            | 0.30%         |
| Exercise Habits            | 25            | 0.25%         |
| Smoking                    | 25            | 0.25%         |
| Family Heart Disease       | 21            | 0.21%         |
| Diabetes                   | 30            | 0.30%         |
| BMI                        | 22            | 0.22%         |
| High Blood Pressure        | 26            | 0.26%         |
| Low HDL Cholesterol        | 25            | 0.25%         |
| High LDL Cholesterol       | 26            | 0.26%         |
| <b>Alcohol Consumption</b> | <b>2586</b>   | <b>25.86%</b> |
| Stress Level               | 22            | 0.22%         |
| Sleep Hours                | 25            | 0.25%         |
| Sugar Consumption          | 30            | 0.30%         |
| Triglyceride Level         | 26            | 0.26%         |
| Fasting Blood Sugar        | ≈29           | ≈0.29%        |
| CRP Level                  | ≈22           | ≈0.22%        |
| Homocysteine Level         | ≈21           | ≈0.21%        |
| Heart Disease Status       | 0             | 0.00%         |

The most notable finding is that **Alcohol Consumption** has a very high missing rate of **25.86%** (2,586 out of 10,000 records). All other features have negligible missingness below 0.31%.

### 3.3 Statistical Summary

Descriptive statistics for the numerical features are summarised below:

**Table 4:** Statistical Summary of Numerical Features

| Feature            | Mean          | Std Dev       | Min   | Max    |
|--------------------|---------------|---------------|-------|--------|
| Age                | 49.30         | 18.19         | 18.0  | 80.0+  |
| Blood Pressure     | 149.76        | 17.57         | 120.0 | 180.0+ |
| Cholesterol Level  | 225.43        | 43.58         | 150.0 | 300.0+ |
| BMI                | 29.08         | 6.31          | 18.0  | 50.0+  |
| Sleep Hours        | $\approx 6.9$ | $\approx 1.7$ | 3     | 12     |
| Triglyceride Level | $\approx 200$ | $\approx 60$  | 50    | 400+   |

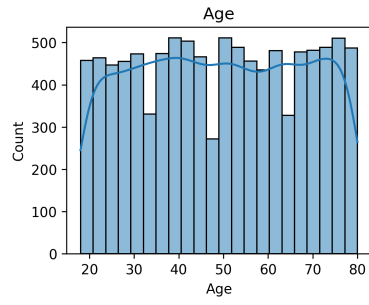
Numerical features show broad ranges, indicating that StandardScaler will be essential before applying a distance-based algorithm such as KNN.

### 3.4 Correlation Analysis

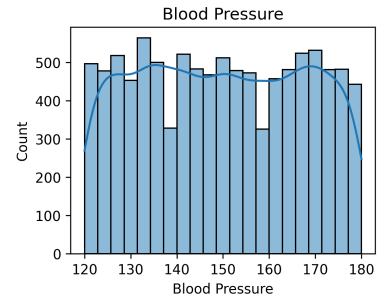
A correlation matrix was computed for all numerical features. The resulting values indicate very low linear correlations among all numerical feature pairs (coefficients below  $\pm 0.03$  in most cases), suggesting that the numerical features are largely independent of one another.

### 3.5 Feature Distribution Plots

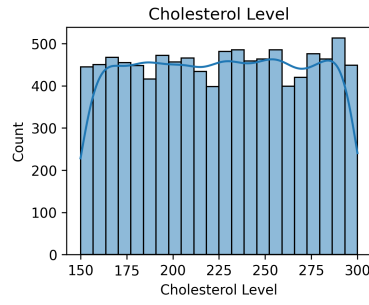
Distribution histograms with kernel density estimates (KDE) were generated for each of the nine numerical features. The figures below show the distributions.



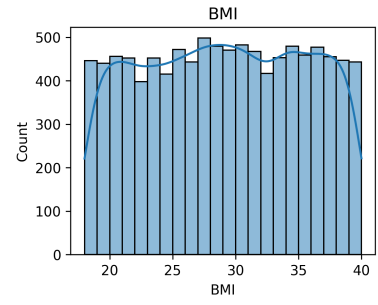
(a) Age



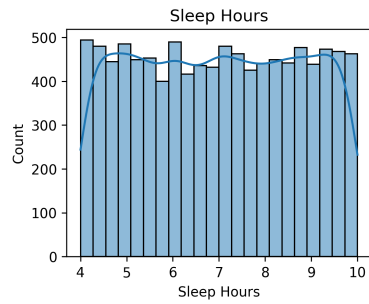
(b) Blood Pressure



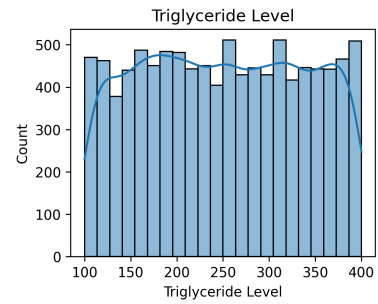
(c) Cholesterol Level



(d) BMI



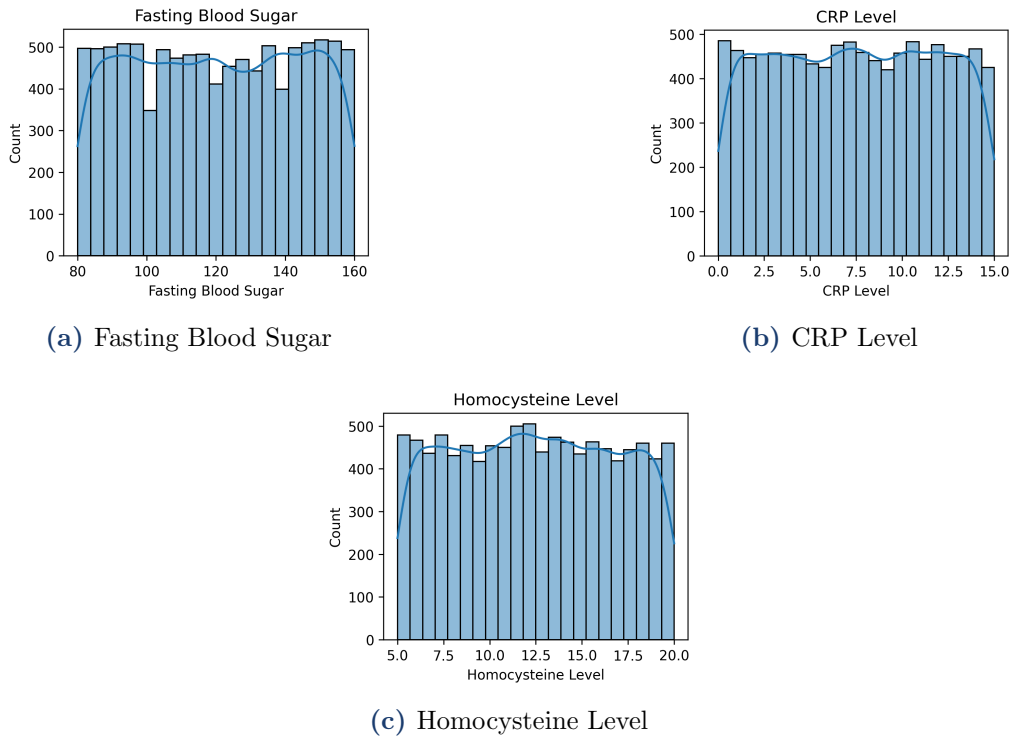
(e) Sleep Hours



(f) Triglyceride Level

**Figure 1:** Distribution plots (with KDE) for numerical features — Part 1





**Figure 2:** Distribution plots (with KDE) for numerical features — Part 2

## 4. Data Preprocessing

### 4.1 Handling Missing Values

Two imputation strategies are applied based on data type:

- **Numerical columns** (9 features): Missing values are filled with the **median** of each column. The median is preferred over the mean as it is robust to outliers commonly present in clinical measurements.
- **Categorical columns** (11 features + target): Missing values are filled with the **mode** (most frequent value) of each column.

After imputation, all 21 columns contain zero missing values.

### 4.2 Feature and Target Separation

The feature matrix **X** is formed by dropping the target column **Heart Disease Status**, and the target vector **y** is extracted as a separate series.

### 4.3 Categorical Encoding

The feature matrix still contains categorical string columns. These are converted to binary indicator columns using **one-hot encoding**. The **drop\_first** option removes one dummy column per category to avoid multicollinearity (the dummy variable trap).

After encoding, the feature matrix expands from 20 columns to a larger numerical matrix.

The target variable is binary-encoded: 'No'  $\rightarrow$  0 and 'Yes'  $\rightarrow$  1.

#### 4.4 Train / Test Split

The dataset is divided into training (80%) and test (20%) subsets using stratified splitting with `random_state=42`. Stratification ensures that the class ratio is preserved in both splits, which is especially important given the 80:20 class imbalance.

**Table 5:** Train / Test Split Summary

| Split        | Total Samples | Class 0 (No HD) | Class 1 (HD)    |
|--------------|---------------|-----------------|-----------------|
| Training Set | 8,000         | $\approx$ 6,400 | $\approx$ 1,600 |
| Test Set     | 2,000         | 1,600           | 400             |
| Total        | 10,000        | $\approx$ 8,000 | $\approx$ 2,000 |

#### 4.5 Feature Scaling

KNN computes distances between instances, making it highly sensitive to feature magnitudes. A `StandardScaler` is fitted on the training set and applied to both the training and test sets:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

This transforms each feature to zero mean and unit variance. Fitting only on the training set prevents data leakage from the test set into the preprocessing step.

## 5. Model: K-Nearest Neighbours

### 5.1 Algorithm Overview

K-Nearest Neighbours (KNN) is a *non-parametric, lazy* learning algorithm. It makes no explicit assumption about the underlying data distribution. During training, the algorithm simply stores all training instances. At inference time, for a query point  $\mathbf{x}_q$ , KNN:

1. Computes the distance from  $\mathbf{x}_q$  to every training point.
2. Identifies the  $k$  training points (neighbours) with the smallest distances.
3. Assigns the majority class label among those  $k$  neighbours to  $\mathbf{x}_q$ .

The default distance metric is **Euclidean distance**:

$$d(\mathbf{x}_q, \mathbf{x}_i) = \sqrt{\sum_{j=1}^n (x_{q,j} - x_{i,j})^2}$$

where  $n$  is the number of features.

## 5.2 Model Configuration

The model is instantiated with  $k = 5$  nearest neighbours. This hyperparameter represents a balance between a very noisy  $k = 1$  decision boundary and an overly smooth boundary produced with large  $k$ .

**Table 6:** KNN Model Configuration

| Hyperparameter               | Value                      |
|------------------------------|----------------------------|
| Number of neighbours ( $k$ ) | 5                          |
| Distance metric              | Euclidean (default)        |
| Weights                      | Uniform (default)          |
| Algorithm                    | Auto (ball_tree / kd_tree) |

## 6. Results and Evaluation

### 6.1 Overall Accuracy

**Test Set Accuracy: 79.00%**

(1,580 correctly classified out of 2,000 test samples)

### 6.2 Classification Report

Table 7 presents the per-class precision, recall, F1-score, and support on the held-out test set.

Table 7: Classification Report — KNN ( $k=5$ )

| Class                | Precision | Recall | F1-Score | Support |
|----------------------|-----------|--------|----------|---------|
| 0 (No Heart Disease) | 0.80      | 0.98   | 0.88     | 1,600   |
| 1 (Heart Disease)    | 0.22      | 0.02   | 0.04     | 400     |
| Accuracy             |           | 0.79   |          | 2,000   |
| Macro Average        | 0.51      | 0.50   | 0.46     | 2,000   |
| Weighted Average     | 0.68      | 0.79   | 0.71     | 2,000   |

6.3 Confusion Matrix

Table 8 shows the confusion matrix, and Figure 3 provides a visual heatmap representation.

Table 8: Confusion Matrix ( $k=5$ , Test Set)

|        |           | Predicted |        |
|--------|-----------|-----------|--------|
|        |           | No HD (0) | HD (1) |
| Actual | No HD (0) | 1,572     | 28     |
|        | HD (1)    | 392       | 8      |

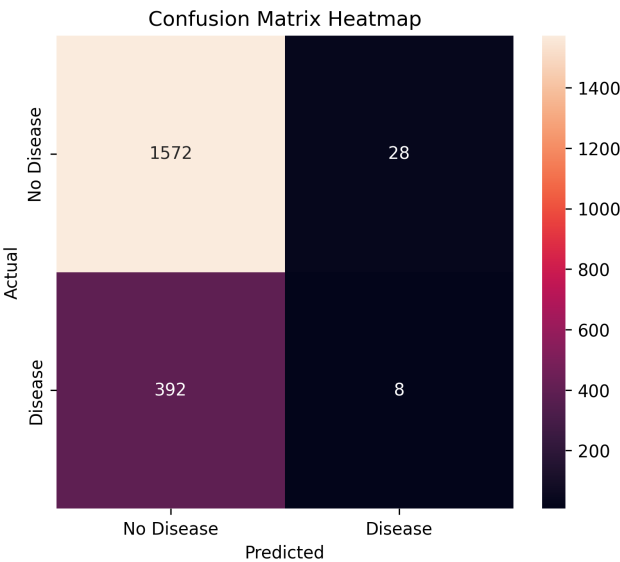


Figure 3: Confusion matrix heatmap — KNN ( $k=5$ ) on the test set

From the confusion matrix:

- **True Negatives (TN):** 1,572 — correctly predicted “No Heart Disease”
- **False Positives (FP):** 28 — healthy patients incorrectly flagged
- **False Negatives (FN):** 392 — heart disease patients *missed*
- **True Positives (TP):** 8 — correctly identified heart disease patients

6.4 Summary of Performance Metrics

**Table 9:** Consolidated Performance Summary

| Metric              | Value      |
|---------------------|------------|
| Overall Accuracy    | 79.00%     |
| Precision (Class 0) | 0.80 (80%) |
| Precision (Class 1) | 0.22 (22%) |
| Recall (Class 0)    | 0.98 (98%) |
| Recall (Class 1)    | 0.02 (2%)  |
| F1-Score (Class 0)  | 0.88       |
| F1-Score (Class 1)  | 0.04       |
| Macro Avg F1        | 0.46       |
| Weighted Avg F1     | 0.71       |

7. Discussion

7.1 Model Performance Analysis

The KNN classifier achieves 79.00% overall accuracy, which appears respectable at first glance. However, a deeper inspection of the per-class metrics reveals a fundamental performance asymmetry. The model performs very well on Class 0 (No Heart Disease) with a recall of 98% and F1-score of 0.88, but performs extremely poorly on Class 1 (Heart Disease), achieving only 2% recall and an F1-score of just 0.04. In practical terms, the model correctly identifies only 8 out of 400 heart disease cases in the test set, missing 392 patients who actually have the disease.

7.2 Class Imbalance

The root cause of this poor minority-class performance is **class imbalance**. The dataset has a 4:1 ratio of healthy to diseased patients (approximately 8,000 vs. 2,000 in the full dataset). When KNN votes among  $k = 5$  nearest neighbours, the dominant class tends to win simply because more of those neighbours belong to the majority class, regardless of the query point’s true label. This makes KNN — and indeed most

standard classifiers without adjustment — poor at detecting minority class samples in imbalanced datasets.

### 7.3 Limitations

Several limitations should be acknowledged:

1. **No class imbalance handling:** Neither SMOTE (oversampling), class weighting, nor threshold adjustment was applied. These techniques could substantially improve Class 1 recall.
2. **Fixed  $k$  value:**  $k = 5$  was used without cross-validated hyperparameter tuning. The optimal  $k$  may differ and should be determined via a grid search or elbow-method analysis.
3. **Alcohol Consumption imputation:** With 25.86% of this column missing, imputing with the mode introduces substantial information loss. Multiple imputation or indicator variables could be more appropriate.
4. **High-dimensional KNN:** After one-hot encoding, the feature space grows considerably. Distance-based models suffer from the “curse of dimensionality” in high dimensions, which can degrade KNN performance.
5. **No feature selection:** The near-zero correlations among numerical features suggest many features may contribute little signal. Feature importance analysis (e.g., using Random Forest permutation importance) was not performed.

### 7.4 Potential Improvements

Future work could investigate:

- Applying **SMOTE** or **class\_weight='balanced'** to address the imbalance problem.
- Tuning  $k$  via cross-validated grid search.
- Comparing KNN against ensemble methods (Random Forest, XGBoost) and logistic regression.
- Applying **PCA** or recursive feature elimination to reduce dimensionality before KNN.
- Evaluating AUC-ROC as a threshold-independent performance metric.

## 8. Conclusion

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This project implemented a complete end-to-end machine learning pipeline for heart disease prediction using the K-Nearest Neighbours classifier on a dataset of 10,000 patient records. The pipeline covered data loading and inspection, exploratory data analysis, median/mode imputation for missing values, one-hot encoding of categorical features, stratified train/test splitting, standard scaling, and KNN model training with  $k = 5$ .

The model achieved an overall test accuracy of 79.00%. While this accuracy looks reasonable, the classification report exposed a severe class imbalance problem: the model correctly identifies only 2% of actual heart disease cases, making it clinically inadequate without further adjustment. This highlights that accuracy alone is a misleading metric for imbalanced binary classification and that recall (sensitivity) for the positive class is the more clinically relevant measure.

The project successfully demonstrates the full scikit-learn pipeline and provides a meaningful baseline upon which further improvements — such as resampling, hyperparameter tuning, dimensionality reduction, and alternative classifiers — can be built to produce a practically useful heart disease screening tool.

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